Lab 7(A) - Laplacian Filter

In [8]:
```
import cv2
import numpy as np

from IPython.display import Image, display
from matplotlib import pyplot as plt

plt.rc('text', usetex=False)
```

Apply Laplacian Filter on Image to get Sharp Image

In [9]:
```
# Manual Laplacian Filter
img = cv2.imread("moon.jpg", 0)

manual_kernel = np.array([
    [0,  1,  0],
    [1, -4,  1],
    [0,  1,  0]
], dtype=np.float32)

laplacian_manual = cv2.filter2D(img, cv2.CV_64F, manual_kernel)
laplacian_abs = cv2.convertScaleAbs(laplacian_manual)

finalImage = np.clip(
    img.astype(np.float64) - laplacian_manual, 0, 255
).astype(np.uint8)

plt.figure(figsize=(12, 8))

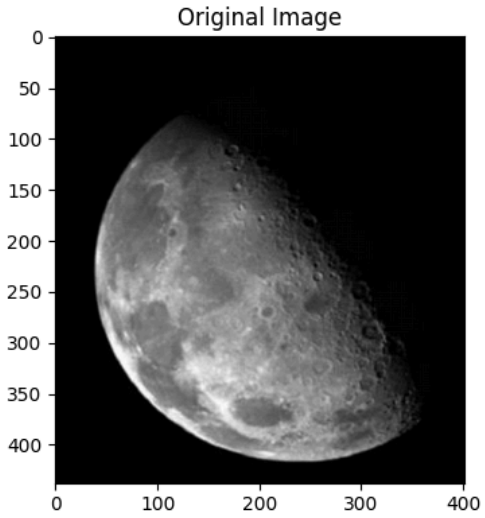
plt.subplot(221)
plt.imshow(img, cmap='gray')
plt.title('Original Image')

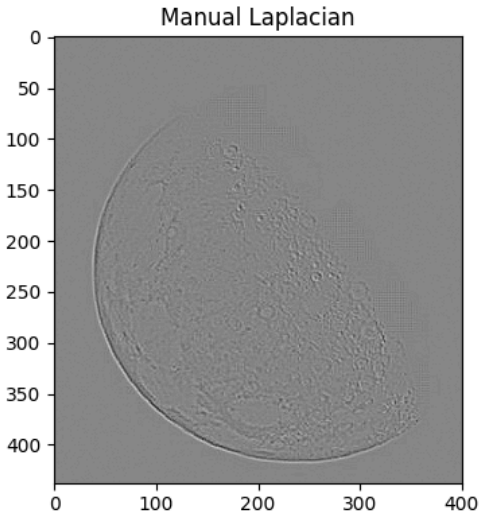
plt.subplot(222)
plt.imshow(laplacian_manual, cmap='gray')
plt.title('Manual Laplacian')

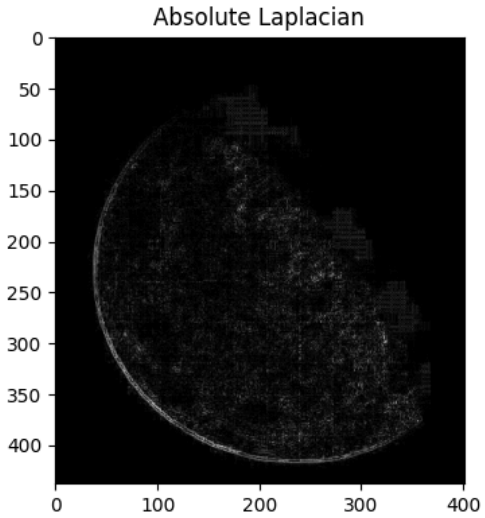
plt.subplot(223)
plt.imshow(laplacian_abs, cmap='gray')
plt.title('Absolute Laplacian')

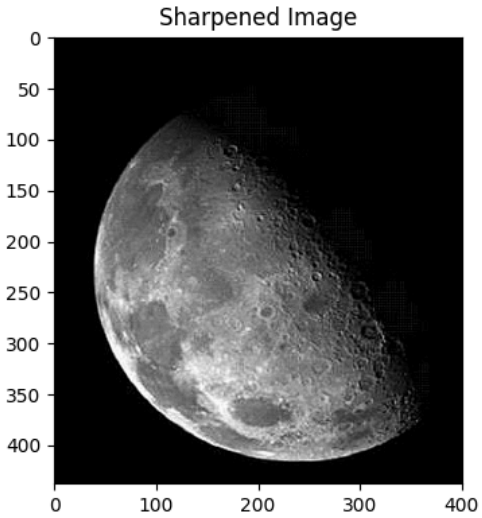
plt.subplot(224)
plt.imshow(finalImage, cmap='gray')
plt.title('Sharpened Image')

plt.tight_layout()
plt.show()
```









In [10]:
```
# OpenCV Laplacian Filter
blurred = cv2.GaussianBlur(img, (3, 3), 0)

laplacian = cv2.Laplacian(blurred, cv2.CV_64F)
laplacian_8bit = cv2.convertScaleAbs(laplacian)

finalImage2 = np.clip(
    blurred.astype(np.float64) - laplacian, 0, 255
).astype(np.uint8)

plt.figure(figsize=(12, 8))

plt.subplot(221)
plt.imshow(img, cmap='gray')
plt.title('Original Image')

plt.subplot(222)
plt.imshow(laplacian, cmap='gray')
plt.title('OpenCV Laplacian')

plt.subplot(223)
plt.imshow(laplacian_8bit, cmap='gray')
plt.title('Absolute Laplacian')
```

```
plt.subplot(224)
plt.imshow(finalImage2, cmap='gray')
plt.title('Sharpened Image')

plt.tight_layout()
plt.show()
```

